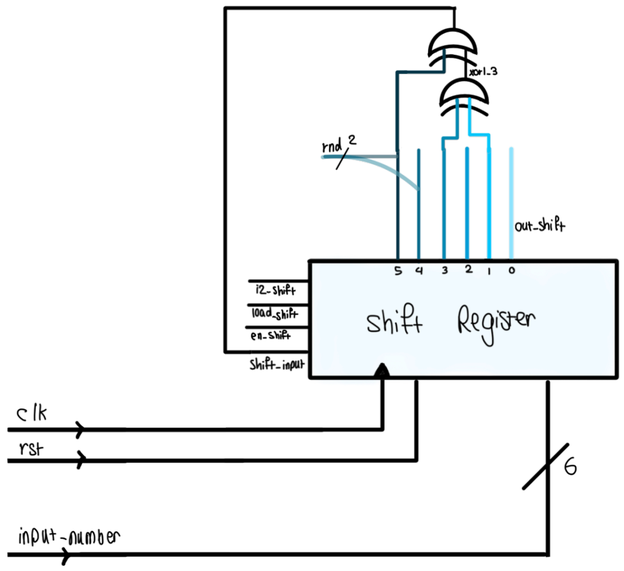
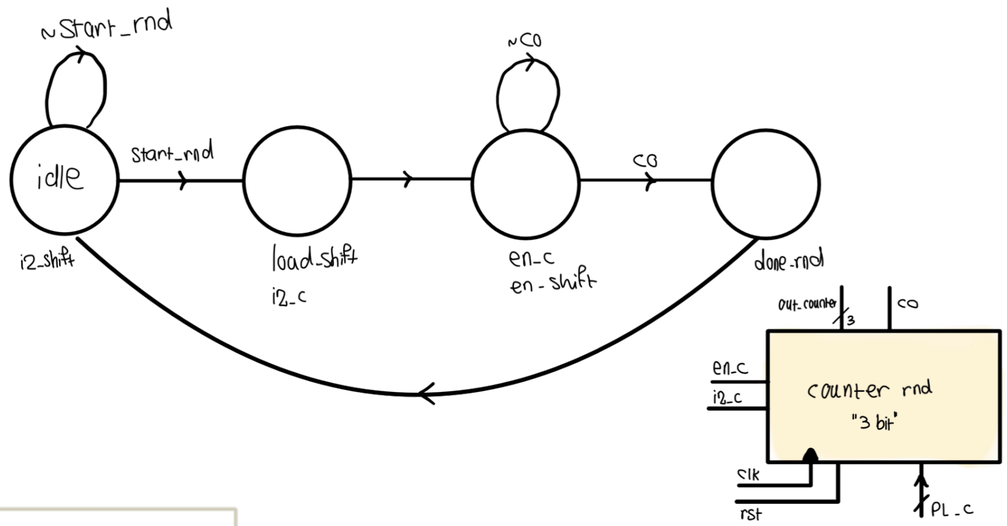
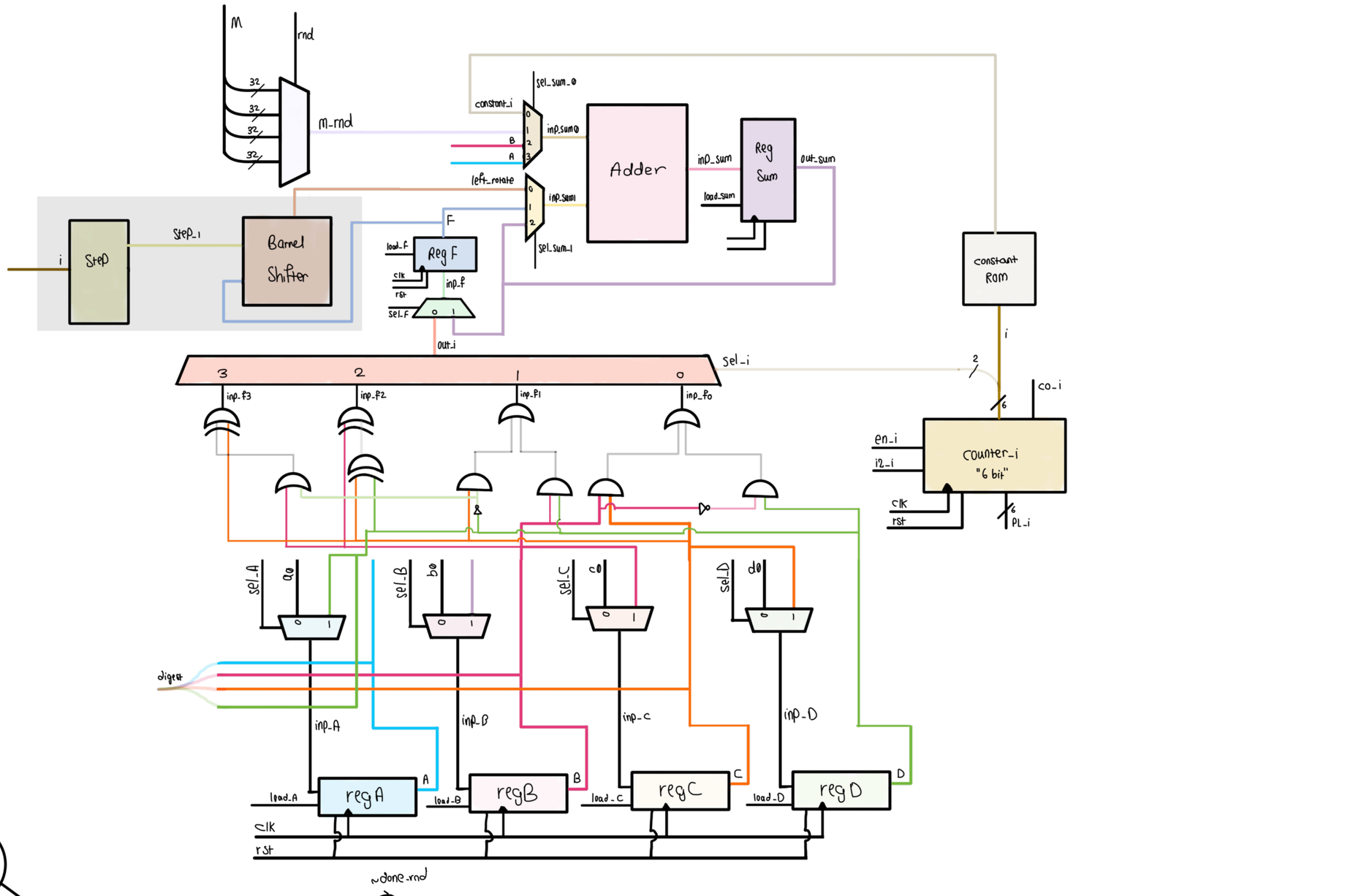
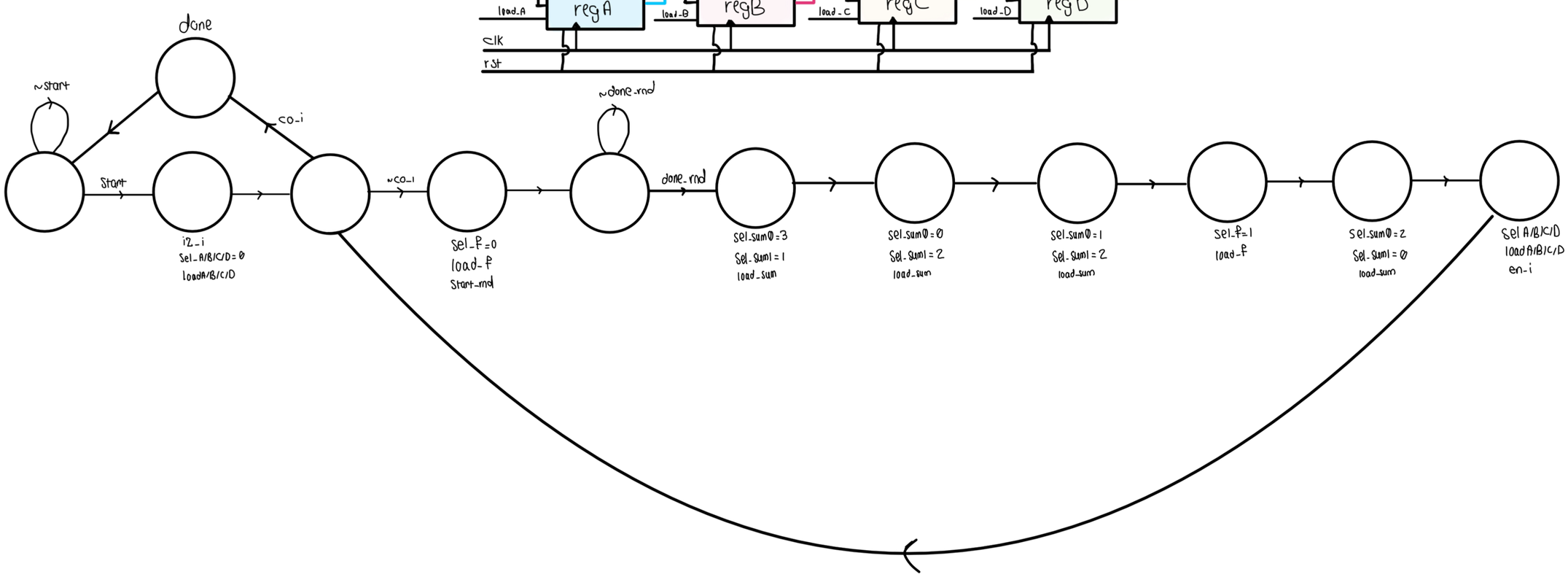

xor1-3
rnd 2
out-shift
5 4 3 2 1 0
i2_shift
load_shift
en_shift
Shift_input
Shift Register
clk
rst
6
input_number


~Start_rnd
idle
Start_rnd
load_shift
i2_c
~CO
en_c
en_shift
CO
done_rnd
i2_shift
out_counter
3
CO
en_c
i2_c
counter rnd
"3 bit"
clk
rst
PL_c


M
rnd
32
32
32
32
M_rnd
constant_i
sel_sum_0
0
1
2
3
B
A
inp_sum0
left_rotate
0
1
2
inp_sum1
sel_sum_1
Adder
inp_sum
load_sum
Reg Sum
out_sum
i
Step
Step_1
Barrel Shifter
F
load_f
clk
rst
Reg F
inp_f
sel_F
0 1
out_i
constant Rom
i
2
6
co_i
en_i
i2_i
counter_i
"6 bit"
clk
rst
PL_i
sel_i
3
2
1
0
inp_f3
inp_f2
inp_f1
inp_f0
&
sel_A
Qa
sel_B
ba
sel_C
ca
sel_D
da
0 1
digit
inp_A
inp_B
inp_C
inp_D
load_A
load_B
load_C
load_D
regA
regB
regC
regD
A
B
C
D
clk
rst


done
~start
Start
i2_i
Sel_A/B/C/D = 0
load A/B/C/D
co_i
~co_i
Sel_F = 0
load_f
Start_rnd
~done_rnd
done_rnd
Sel_sum0 = 3
Sel_sum1 = 1
load_sum
Sel_sum0 = 0
Sel_sum1 = 2
load_sum
Sel_sum0 = 1
Sel_sum1 = 2
load_sum
Sel_F = 1
load_f
Sel_sum0 = 2
Sel_sum1 = 0
load_sum
Sel A/B/C/D
load A/B/C/D
en_i